Regularization → Linear
Non-linear

① Bias-variance tradeoff!

② Feature selection!

③ Penalization!

Bias

β^*

↓

data → $Est(\beta)$

$|\beta^* - Est(\beta)|$

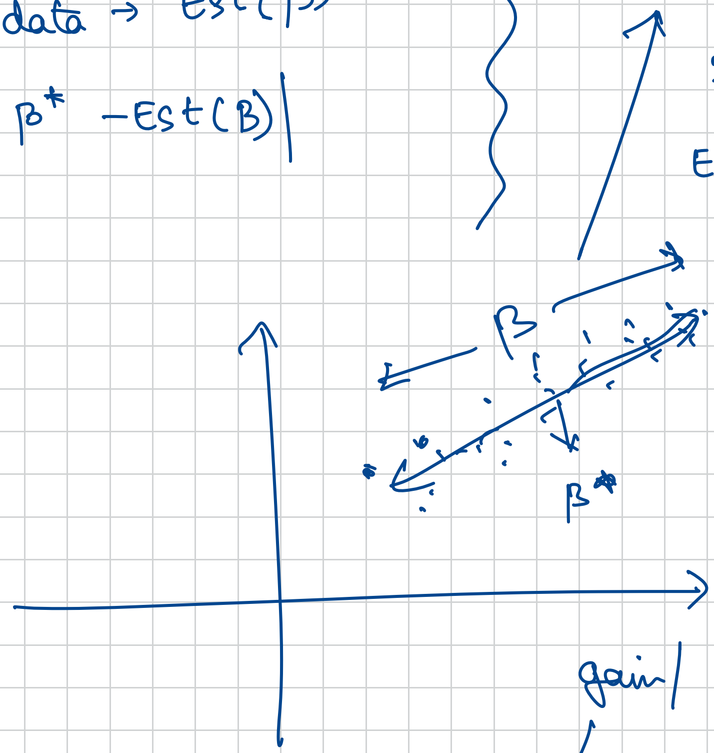
Variance

1, 2, 3, 4, 5

$E(N) = 15/5 = 3$

2, 3, 3, 4, 12

$E(N) = 7\cancel{8} = 6$



→ parameters

↓

weights

↓

Outcomes...

gain / resource

$$y = \omega_1 x_1 + \omega_2 x_2$$

$\omega_2 >> \omega_1$

$$J(\theta) = \| [X]\vec{\theta} - \vec{y} \|^2$$

$$J(\theta) = (X\vec{\theta} - \vec{y})^T (X\vec{\theta} - \vec{y})$$

$$J(\theta) = \theta^T X^T X \theta - \vec{\theta}^T X^T y - y^T X \theta + y^T y$$

$$\frac{\partial J}{\partial \theta} = 0 \Rightarrow \boxed{\hat{\theta} = (X^T X)^{-1} X^T y}$$

$\underline{X^T}\,\underline{X}$?

↓

$X^T X$: square; symmetric

$$\vec{X} = \left[\begin{array}{cccc} \text{cloud} & \text{Tin} & & \\ | & | & & | \\ x_1 & x_2 & \cdots & x_n \\ | & | & & | \end{array}\right]_{m \times n}$$

↗ # of samples

$$X^T X = \underbrace{Q}_{\text{collection of eigenvectors}} \overset{\text{diagonal}}{\Lambda} Q^T$$

$$\begin{bmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_n \end{bmatrix}$$

$$(X^T X)^{-1} = \left(\frac{1}{\Lambda}\right)$$

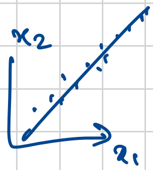
$$QQ^T = I$$

$$\|q_j\| = 1$$

$$q_j^T q_i = 0 \quad \forall\ i \neq j$$

x_1 & x_2 are strongly correlated

x_1 y x_2



$R^2 = 1.0$

Collinearity

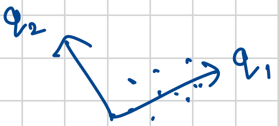
$$\begin{cases} y = \alpha_1 x_1 + \alpha_2 x_2 \\ y = \underbrace{(\alpha_1 + \alpha_2)}\,(x_1) \end{cases}$$

$\alpha_1 + \alpha_2 = 8$

$+10000\ ;\ \alpha_2 = -9992$

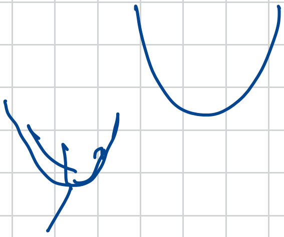
$$\left.\begin{array}{l} \hat{\beta} = (X^T X)^{-1} \\ \Delta \hat{\beta} \propto (X^T X)^{-1} \end{array}\right\} \text{Linearity}$$

PCA →



$$(X^T X^{-1})^{-1} \propto \frac{1}{\nu_j}$$

$\nu_j \to 0$



Penalties

$$J(\vec{\theta}) = \|X\theta - y\|^2 \Big\}\ \underline{OLS}$$

"hyperparameter"

penalty

Parsimony

$$J(\vec{\theta}) = \|X\theta - y\|^2 + \lambda \|\theta\|^2$$

$$J(\theta) = \underline{(X\theta - y)^T (X\theta - y)} + \underline{\lambda \theta^T \theta}$$

$$J(\theta) = \theta^T X^T X \theta - y^T X \theta - \theta^T X^T y + y^T y + \lambda \theta^T \theta$$

$$\frac{\partial J(\theta)}{\partial \theta} = 0$$

$$\hat{\theta} = \left(X^T X + \lambda I \right)^{-1} X^T y$$

$$A q = \nu q$$

$$(X^T X + \lambda I) q_j = \boxed{X^T X q_j} + \lambda I a_j$$

$$= \mu_j q_j + \lambda a_j$$

$$(X^T X + \lambda I) q_j = \boxed{(\nu_j + \lambda)} q_j$$

$$\hat{\beta}_{OLS} = \left(X^T X \right)^{-1} X^T y$$

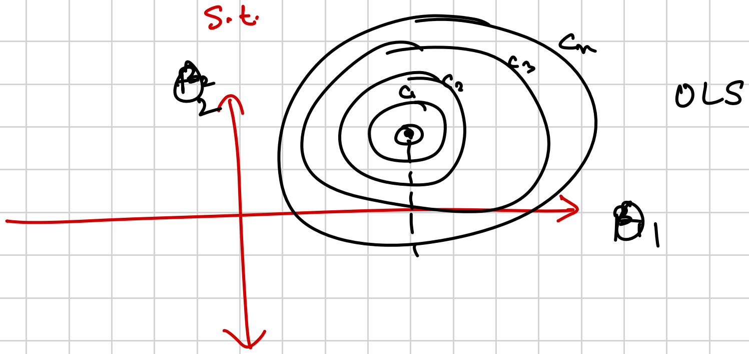
$$\propto \frac{1}{\nu}$$

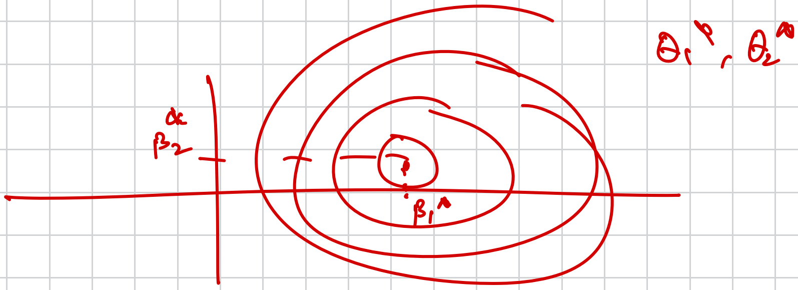
$$\hat{\beta}_{\text{shrinkg}} \propto \frac{1}{\lambda + \nu}$$

$$\| \beta \|^2$$

$$\beta_1^2 + \beta_2^2 = \frac{1}{\lambda} \quad \text{Circle} \qquad t$$

S.t.

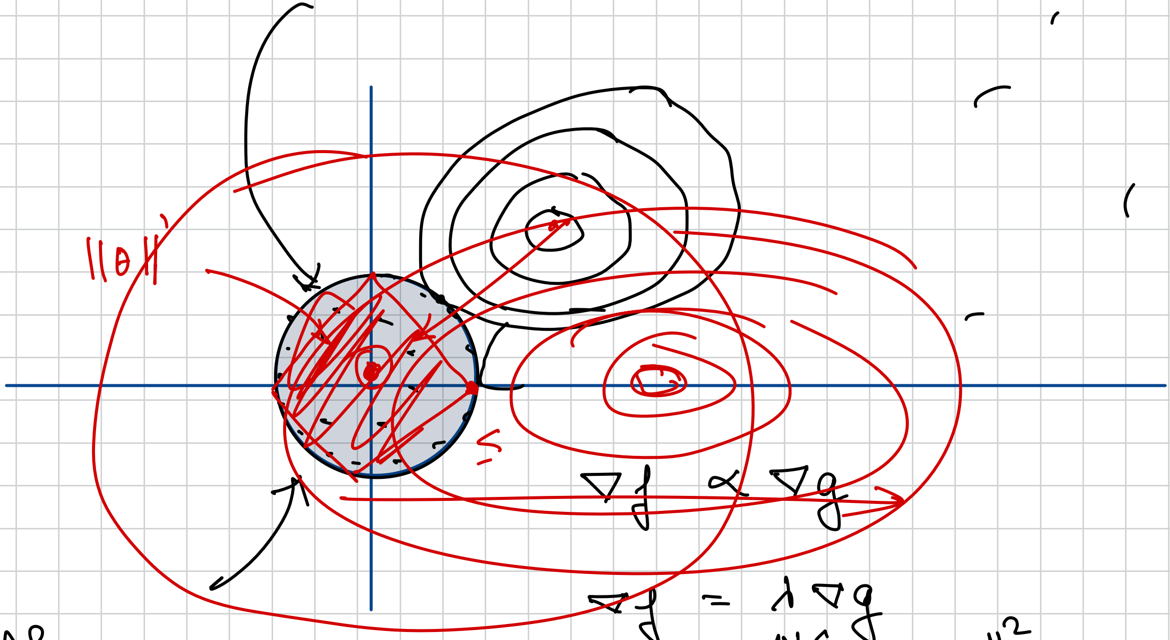

θ_2
c_n
c_3
c_2
c_1
OLS
θ_1


θ_1^*, θ_2^*
β_2^*
β_1^*

objective min $= \|X\theta - y\|^2$ error

s.t.

$\|\theta\|^2 \leq t$ smaller t: $\lambda \uparrow$


$\|\theta\|^2$
$\leq$
$\nabla f \propto \nabla g$

$\nabla f = \lambda \nabla g$

Biasing the parameter

$$\mathcal{L} = \overset{\text{OLS}}{f} + \lambda \overset{\|w\|^2}{g}$$